

Introduction to Machine Learning

Lab File Link: [🔗 Introduction to ML](#)

1. Introduction

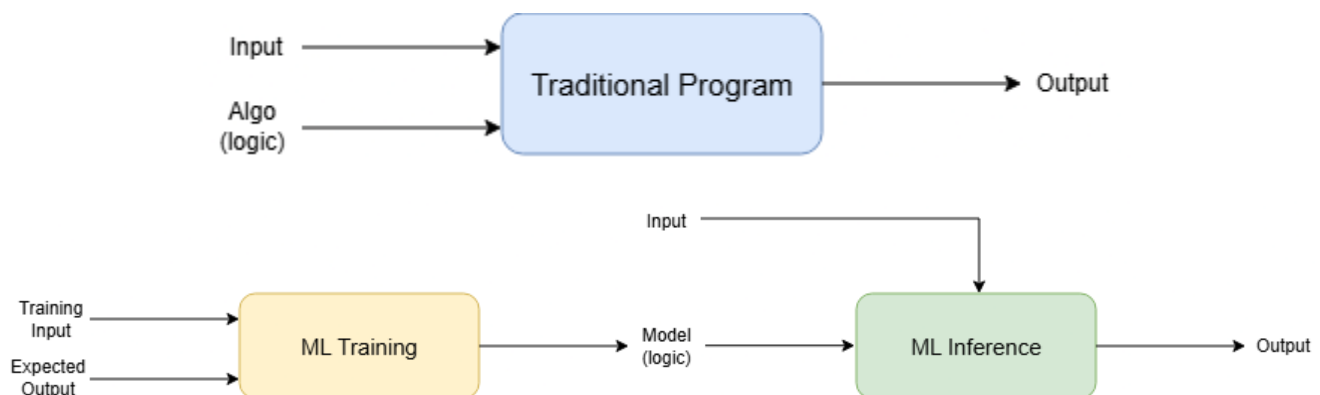
Machine learning (ML) focuses on developing algorithms and models that enable computers to **learn patterns, relationships, and structures from data and make decisions or predictions based on new data without being explicitly programmed for specific tasks.**

ML handles unpredictable situations by learning patterns from past data and making generalizations. Some tasks involve complex patterns and uncertainty, such as face recognition. These are situations where ML often works better by learning from data rather than following fixed instructions.

2. DS vs AI vs ML vs DL

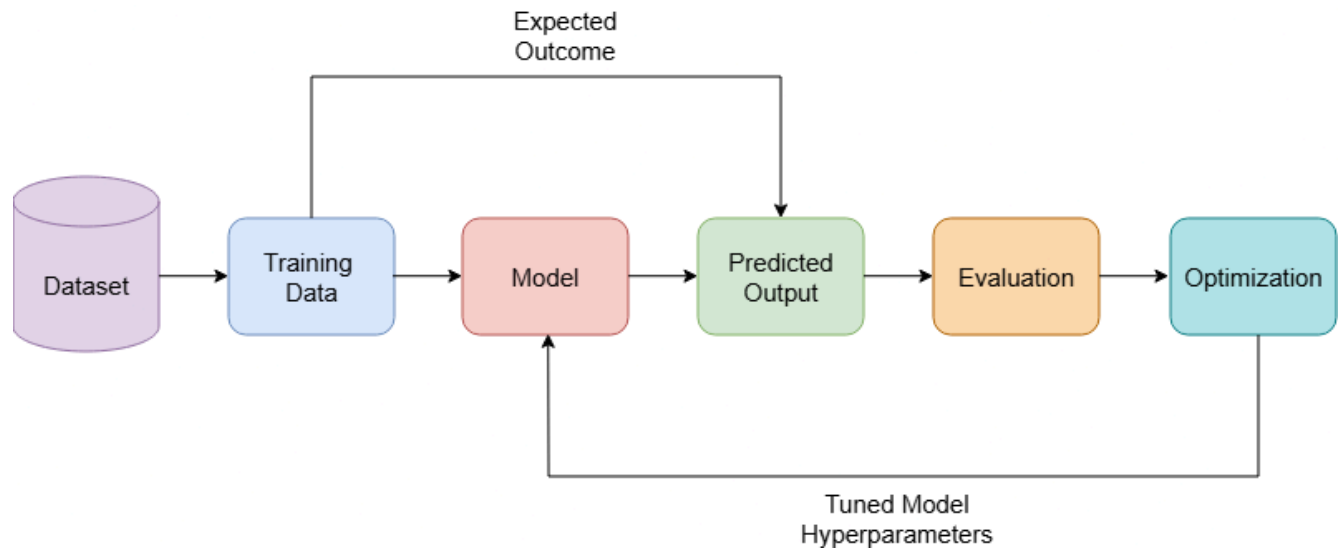
- **Data Science (DS):** Uses statistical methods, algorithms, and technology to extract insights and knowledge from structured and unstructured data.
- **Artificial Intelligence (AI):** Simulation of human intelligence in machines, enabling them to perform tasks that typically require human cognition.
- **Machine Learning (ML):** A subset of AI focused on developing algorithms that allow computers to learn from data and make decisions without being explicitly programmed.
- **Deep Learning (DL):** A subset of ML that uses neural networks with many layers to model and solve complex patterns and relationships in data.

Visual Comparison: Traditional Programming vs Machine Learning



3. Generic ML Pipeline

A generic machine learning pipeline starts with a **dataset**, prepares the training data, builds a **model**, makes **predictions**, evaluates performance, and then improves the model through **optimization and parameter tuning**.

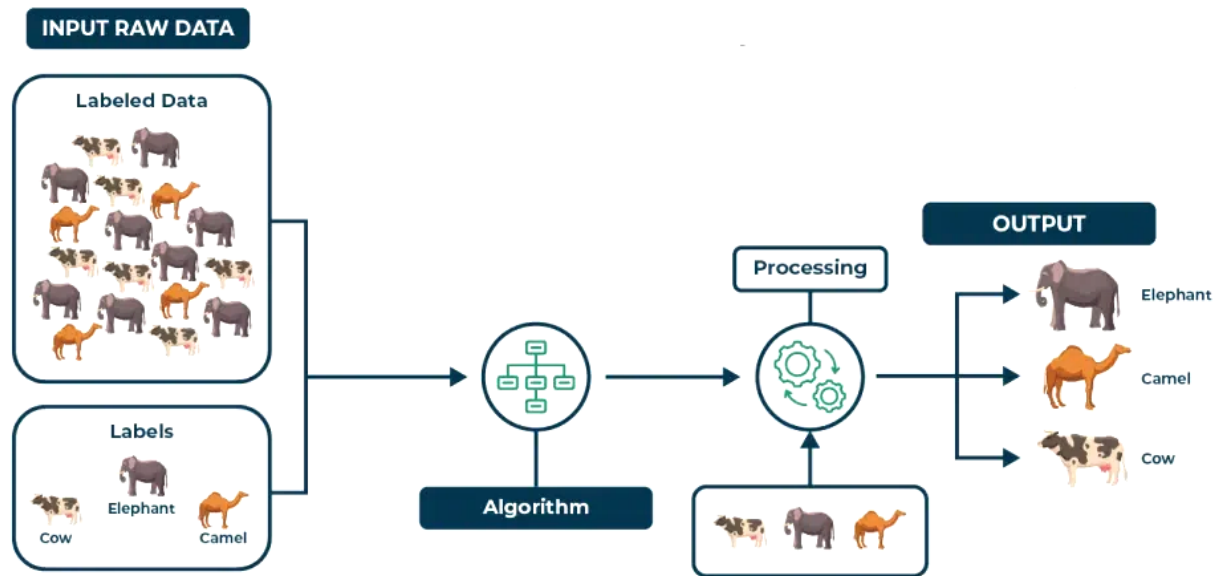


4. Three Major Types of Machine Learning

1. **Supervised Learning:** Uses labeled data to train models, where each input has a known output. It is used to predict outcomes for new data.
2. **Unsupervised Learning:** Works with unlabeled data, helping to find hidden patterns such as clusters.
3. **Reinforcement Learning:** Involves an agent interacting with an environment and receiving rewards or penalties.

5. Supervised Learning

In supervised learning, **the model learns from input-output pairs, also called feature-label pairs**. The goal is for the model to **learn a mapping from inputs to outputs so it can accurately predict outcomes for new, unseen data**.



Key Terms

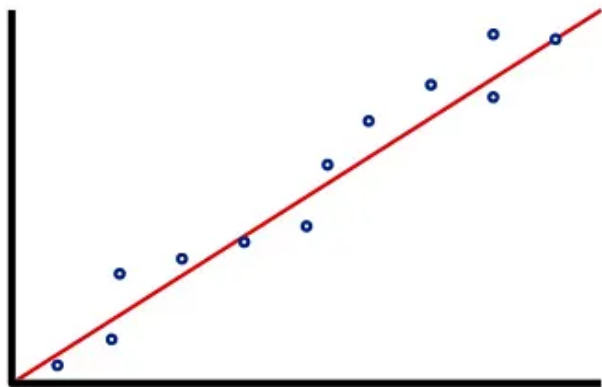
- **Input (Features):** Measurable variables that form the input vector X . Example: for predicting house prices, features can include house size, number of bedrooms, location rating, and age of the house.
- **Output (Label / Target):** The value or category to be predicted, usually denoted by Y . Example: the price of a house in dollars.

Steps of Supervised Learning

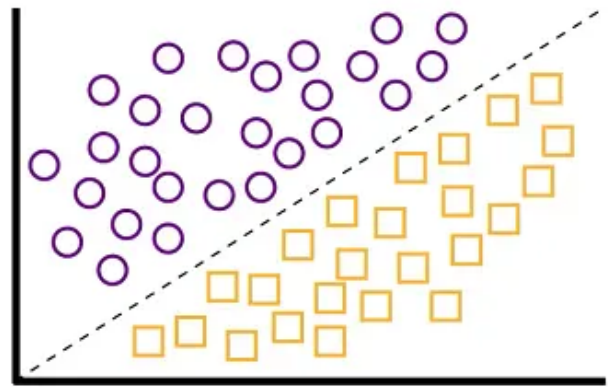
1. **Collect Data:** Gather labeled data where each input has a corresponding correct output.
2. **Preprocess Data:** Clean the data, handle missing values, normalize features, and encode categorical variables if necessary.
3. **Feature Engineering:** Create new features or select the most relevant features to improve model performance.
4. **Split the Data:** Divide the dataset into training and test sets, and optionally a validation set.
5. **Train the Model:** Use the training set to teach the model to learn the mapping from input to output.
6. **Evaluate the Model:** Use the test set to measure performance using appropriate metrics.
7. **Tune Hyperparameters:** Adjust model settings to optimize performance.
8. **Deploy the Model:** Use the trained model to make predictions on new, unseen data.

Types of Supervised Learning

- **Regression:** Predicts continuous numeric values. Example: predicting house prices based on size and location.
- **Classification:** Predicts discrete class labels. Example: classifying emails as "spam" or "not spam".



Regression



Classification

6. Hypothesis in Supervised Learning

A hypothesis is a candidate function that maps inputs to outputs.

$$h : X \rightarrow Y$$

where X = set of features, Y = set of labels

The goal is to select the best hypothesis from the hypothesis space through a loss function, model training, and validation.

7. Bias, Underfitting, Variance, and Overfitting

- **Bias** is the error due to overly strong assumptions in the model. **Underfitting** happens when a model is too simple to capture the underlying patterns. It usually leads to poor performance on both training and test data.
- **Variance** is the error due to excessive sensitivity to small fluctuations in the training data. A high-variance model captures noise along with the signal. **Overfitting** happens when a model is too complex and fits the noise in the training data. High variance often gives excellent performance on training data but poor generalization to new data.

A good fit in machine learning is a **model that captures the underlying structure of the data without memorizing noise**. It strikes a balance between underfitting and overfitting.